

RMO(ORISSA) – 2002

Max Mark – 100

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer as many questions as you can.
- Q. 1 to Q.10 carry 10 marks each.
- Bonus question will be taken into account in case of tie.

1.

(a) In a running competition of d metre, A lagging behind B by 20 metre, B lagging behind C by 10 metre and A lagging behind C by 28 metre. Find the value of d .

(b) One shopkeeper purchases an item of costing Rs 24 by $12\frac{1}{2}\%$ less. Shopkeeper after giving 20%

discount on list price of that item got a profit of $33\frac{1}{3}\%$. Find the list price of the item.

2.

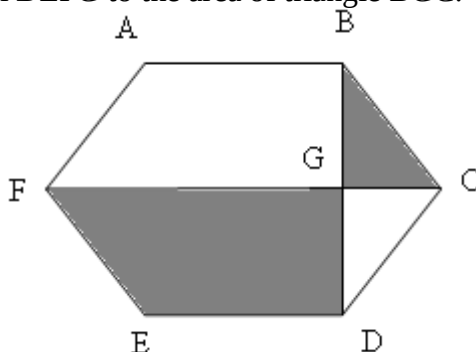
(a) $n! = 1 \times 2 \times 3 \times \dots \times n$. If $n! = 2^{15} \times 3^6 \times 5^3 \times 7^2 \times 11 \times 13$, then find value of n .

(b) Find the remainder when $2000^{2002} + 2002^{2000}$ divided by 2003.

3.

(a) In a quadrilateral ABCD, E and F are two points on BC such that E is more nearer to B than F. If $m\angle BAE = m\angle CDF$ and $m\angle EAF = m\angle FDE$, then prove that $m\angle FAC = m\angle EDB$.

(b) In the following regular hexagon ABCDEF, two diagonals FC and BD intersect at G. Find the ratio of area of quadrilateral DEFG to the area of triangle BGC.



4.

(a) Starting from 1 some consecutive integers are written on a blackboard. After deleting one number out of them the remaining have the average $35\frac{7}{17}$. Then find which number deleted from the sequence of integers.

(b) Find the sum $\sum_{k=0}^n \frac{(-1)^{\left[\frac{3^k}{2}\right]}}{3^k}$, where $[x]$ is the greatest integer less than or equals to x .

5. Inside a regular hexagon of side $2a$ each 25 points are there. Prove that exist At least two points whose at most distance is a .

6. In an isosceles triangle ABC, $AB = BC$. From the circum centre of triangle ABC, a perpendicular is drawn to the CD (which is the angle bisector of $\angle C$), which intersects AB at F. A line parallel to CD passes through E, which intersects AB at F. Prove that $BE = FD$.
7. If $x + y + z = 1$, $x^2 + y^2 + z^2 = 2$, $x^3 + y^3 + z^3 = 3$, find value of $x^4 + y^4 + z^4 = ?$
8. If a, b, c are positive numbers, then prove that $\frac{1}{a^3 + b^3 + abc} + \frac{1}{b^3 + c^3 + abc} + \frac{1}{c^3 + a^3 + abc} \leq \frac{1}{abc}$.
9. If f is a function on all positive integers such that $f(1) + f(2) + \dots + f(n) = n^2 f(n)$ where $n > 1$ then find the value of $f(2002)$.
10. In triangle ABC, O is the circum centre and G is the centroid. R and r be the radius of incircle and circum circle of triangle ABC. Then prove $OG \leq \sqrt{R(R - 2r)}$.

(BONUS QUESTION)

11. One Theorem is given with proof. Read it carefully, write is it true or not with explanation.

Theorem: -All horses are of same color.

Proof: - Let H be the set of all horses.

Let us use principle of mathematical induction to prove this.

For $n = 1$, set H having one horse. For this, no need to prove anything.

One horse is of same color. So $P(1)$ is true.

Let H be a set of $n - 1$ horses of same color.

Now let us take set H with n horses.

Horses are $h_1, h_2, \dots, h_n$

If we remove h_1 from H that set becomes H_1

Similarly removing h_2 from H that set becomes H_2

Hence removing h_n from H that set becomes H_n .

$H_1, H_2, \dots, H_n$ each contains $n - 1$ number of horses of same color.

Let all the horses of H_1 are of color C_1 and all the horses of H_2 are of color C_2 .

But horse h_n is in H_1 and H_2 . So $C_1 = C_2$.

So, H having n horses are of same color.

So, by principle of mathematical induction set having any number of horses are of same color.

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